Heron's Formula

The big idea

To find the area of a triangle with ½ × base × height, you need the height. But what if you only know the three sides, as with a triangular park? **Heron's formula** (from Heron of Alexandria) gives the area from the sides alone.

HERON'S FORMULA

Area = √[s(s − a)(s − b)(s − c)]

where a, b, c are the sides and s = (a + b + c)/2 is the semi-perimeter

FIND THE AREA OF A TRIANGLE WITH SIDES 40 M, 32 M AND 24 M

$s = (40 + 32 + 24)/2 = 48$.

Area = $\sqrt{[48 \times 8 \times 16 \times 24]} = \sqrt{147456} =$ **384 m²**.

Check: $24^2 + 32^2 = 40^2$, so it's right-angled; ½ × 24 × 32 = 384 ✓

AN EQUILATERAL TRIANGLE HAS SIDE 10 CM. FIND ITS AREA.

$s = 15$. Area = $\sqrt{(15 \times 5 \times 5 \times 5)} = 25\sqrt{3} \approx$ **43.3 cm²**.

This matches the formula for an equilateral triangle: $(\sqrt{3}/4)\, a^2$.

Areas of quadrilaterals

Split the quadrilateral into two triangles with a diagonal and use Heron's formula for each.

QUADRILATERAL ABCD WITH AB = 3, BC = 4, CD = 4, DA = 5 CM AND DIAGONAL AC = 5 CM

△ABC (3, 4, 5) is right-angled: area = ½ × 3 × 4 = 6 cm².

△ACD (5, 4, 5): s = 7, area = √(7 × 2 × 3 × 2) = √84 ≈ 9.2 cm².

Total ≈ **15.2 cm²**.

FACTORISE BEFORE MULTIPLYING

Under the root, break numbers into primes and pair them up. √(48 × 8 × 16 × 24) is easier as $\sqrt{(2^{14} \times 3^2)} = 2^7 \times 3 = 384$.

FORGETTING TO HALVE THE PERIMETER

s is the **semi**-perimeter. Using the full perimeter gives a wrong area.

Quick recap

- Area = √[s(s – a)(s – b)(s – c)], s = half the perimeter.
- Works for any triangle; no height needed.
- Quadrilaterals: split into two triangles.

Practice questions

1. The semi-perimeter of a triangle with sides 5, 12 and 13 cm is
   (a) 30 cm (b) 15 cm (c) 10 cm (d) 60 cm
2. The area of a triangle with sides 5, 12 and 13 cm is
   (a) 60 cm² (b) 30 cm² (c) 65 cm² (d) 78 cm²
3. The area of an equilateral triangle of side 4 cm is
   (a) 4√3 cm² (b) 8√3 cm² (c) 16 cm² (d) 2√3 cm²
4. Heron's formula needs
   (a) base and height (b) the three sides (c) two sides and an angle (d) the angles only
5. To find the area of a quadrilateral with Heron's formula, we
   (a) use the formula directly (b) divide it into two triangles using a diagonal (c) multiply length by breadth (d) use its perimeter only

Answers: 1. b 2. b 3. a 4. b 5. b